

Step 1 — Setting Up Environment

- Install Java 11

```
# apt install openjdk-11-jdk
```

- Check Java version

```
# java -version
```

openjdk version "11.0.8" 2020-07-14

- Download Tomcat from source

```
# cd /opt
```

```
# wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.37/bin/apache-tomcat-9.0.37.tar.gz
```

- Extract the file

```
# tar -xvzf apache-tomcat-9.0.37.tar.gz
```

```
# sudo mv apache-tomcat-9.0.37 tomcat
```

- Create Tomcat User & change ownership

```
# sudo useradd tomcat
```

```
# sudo chown -R tomcat:tomcat tomcat
```

```
# sudo chmod -R 755 tomcat
```

- Make the service run as a service

```
# sudo vi /etc/systemd/system/tomcat.service
```

[Unit]

Description=Tomcat 9 servlet container

After=network.target

[Service]

Type=forking

User=tomcat

Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64"

Environment="JAVA_OPTS=-Djava.security.egd=file:///dev/urandom -Djava.awt.headless=true"

Environment="CATALINA_BASE=/etc/tomcat"

Environment="CATALINA_HOME=/etc/tomcat"

Environment="CATALINA_PID=/etc/tomcat/temp/tomcat.pid"

Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"

ExecStart=/opt/tomcat/bin/startup.sh

ExecStop=/opt/tomcat/bin/shutdown.sh

[Install]

WantedBy=multi-user.target

- **Restart the daemon**

```
# sudo systemctl daemon-reload
```

- **Start the Tomcat**

```
# sudo systemctl start tomcat
```

Step 2 — Create a Java Application

- **Install a Java build tool**

```
# sudo yum install maven -y
```

- **Check Maven version**

```
# mvn -version
```

Apache Maven 3.6.3

- **Generate project**

```
# mvn archetype:generate -DgroupId=com.app.example
```

```
-DartifactId=java-app -DarchetypeArtifactId=maven-archetype-quickstart
```

```
-DinteractiveMode=false
```

The folder structure should be like this:

```
java-app/  
├── pom.xml  
└── src  
    ├── main  
    │   ├── java  
    │   │   ├── com  
    │   │   │   ├── app  
    │   │   │   │   ├── example  
    │   │   │   │   │   App.java  
    │   ├── test  
    │   ├── java  
    │   │   ├── com  
    │   │   │   ├── app  
    │   │   │   │   ├── example  
    │   │   │   │   │   AppTest.java
```

- **Build the application**

```
# mvn package
```

- **Copy application to Tomcat directory**

```
# cp java-app/target/hello-world-1.war /opt/tomcat/webapps/
```

```
# systemctl restart tomcat
```

- **Check the service**

Coding4Deep